HealthData@IE – setting up health data access body services in Ireland

Deliverable 8.2

PILOT REPORT

INFORMATION MANAGEMENT MATURITY ASSESSMENT TOOL

January 2026

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About Health Information and Quality Authority (HIQA)

The Health Information and Quality Authority (HIQA) is an independent statutory body with responsibility for setting standards for all aspects of health information and monitoring compliance against those standards, as set out in Section 8(1) of the Health Act 2007. Under the Act, HIQA is also charged with evaluating the quality of the information available on health and social care and making recommendations to the Minister and the Health Service Executive (HSE) in relation to improving the health information system.

Conflicts of Interest

None reported.

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1. Background

1.1 The European Health Data Space and HealthData@IE

The European Health Data Space (EHDS) is the first of 14 common EU-wide data spaces established under the European Data Strategy to promote secure and trustworthy data use.[1,2] Supported by the 2022 Data Governance Act and 2024 Data Act, the EHDS aims to ensure citizens can control their health data while enabling its secondary use for research, innovation and policy-making.[3,4] In Ireland, the HealthData@IE project, led by the Department of Health in collaboration with the Health Information and Quality Authority (HIQA), the Health Research Board (HRB) and other key stakeholders, supports the creation of Health Data Access Bodies (HDABs) and the national infrastructure required for data access.[5]

The project focuses on developing:

- a Data Access Application Management System (DAAMs) to receive, track and process applications and to issue permits
- a National Health Dataset Catalogue (nHDC)[6] to facilitate data discovery
- a Secure Processing Environment (SPE) to ensure the secure processing of health data
- workstreams on data quality enhancement to improve the quality of health data made available by a HDAB service
- a structured programme of training and engagement with data users, data holders and the public on the EHDS.

1.2 Overview of Work Package 8: Data quality enhancement

Work Package 8 of the HealthData@IE project, coordinated by HIQA, comprises a programme of work centred on enhancing the quality of data that will be made available through the HDAB and enabling data holders to improve and enhance the data quality. The programme of work emphasises supporting data holders around the implementation of the EHDS and the establishment of the HDAB in Ireland, and ensuring data holders can meet their obligations relating to the secondary use of data.

1.3 Overview of the IMMAT

One of the key tasks of Work Package 8 is to develop and pilot a self-assessment tool to enable data holders to self-assess their level of information management maturity. A maturity framework and associated self-assessment tool (Information Management Maturity Assessment Tool, IMMAT) has been developed in consultation with key stakeholders within the health and social care sector in Ireland, including members of the project Steering Group chaired by the Department of Health (see **Appendix A** for a list of the organisations represented) and the Work Package 8 Working Group (see **Appendix B** for full membership details).

The IMMAT is aligned with the four principles and 12 standards outlined in the *National Standards for Information Management in Health and Social Care* [7] as summarised in Table 1 and supports the implementation of these national standards. The aim of the national standards is to contribute to safer better care by improving the management of health and social care information. These national standards developed by HIQA, and approved and mandated by the Minister for Health, are a set of high-level outcomes that describe how organisations and services can improve practices to achieve safer better care. They are evidence-based and informed by the outcomes of engagement with those who use and provide health and social care services. These national standards act as the cornerstone for preparing the system for digitalisation and the implementation of the EHDS, including the effective sharing of data for secondary purposes to meet the requirements of the EHDS Regulation. As part of the development of these national standards, HIQA has also developed implementation support tools including a self-assessment tool (SAT)[8] for data holders to assess the extent to which they comply with the national standards and identify areas of good practice and where improvement is needed, which should be subsequently outlined in a detailed improvement plan. The SAT and improvement plan should be used on an ongoing basis to identify changes required at an organisation level.

Table 1. *National Standards for Information Management in Health and Social Care*

Principle	Standard	Description
Principle 1. A human rights-based approach	Standard 1.1	Uphold people's rights relating to information
	Standard 1.2	Protect privacy and confidentiality
Principle 2. Safety and wellbeing	Standard 2.1	Optimise the accessibility, use and value of information
	Standard 2.2	Undertake effective stakeholder engagement
Principle 3. Responsiveness	Standard 3.1	Align with best practice regarding data standards and agreed definitions
	Standard 3.2	Enhance data quality
	Standard 3.3	Ensure data security
	Standard 3.4	Develop staff capability and capacity for information management
Principle 4. Accountability	Standard 4.1	Develop strong organisational governance, leadership and management
	Standard 4.2	Implement strategy for information management
	Standard 4.3	Promote effective performance assurance and risk management
	Standard 4.4	Ensure compliance with relevant legislation and codes of practice

From this point on in the report, the *National Standards for Information Management in Health and Social Care* will be referred to as the national standards.

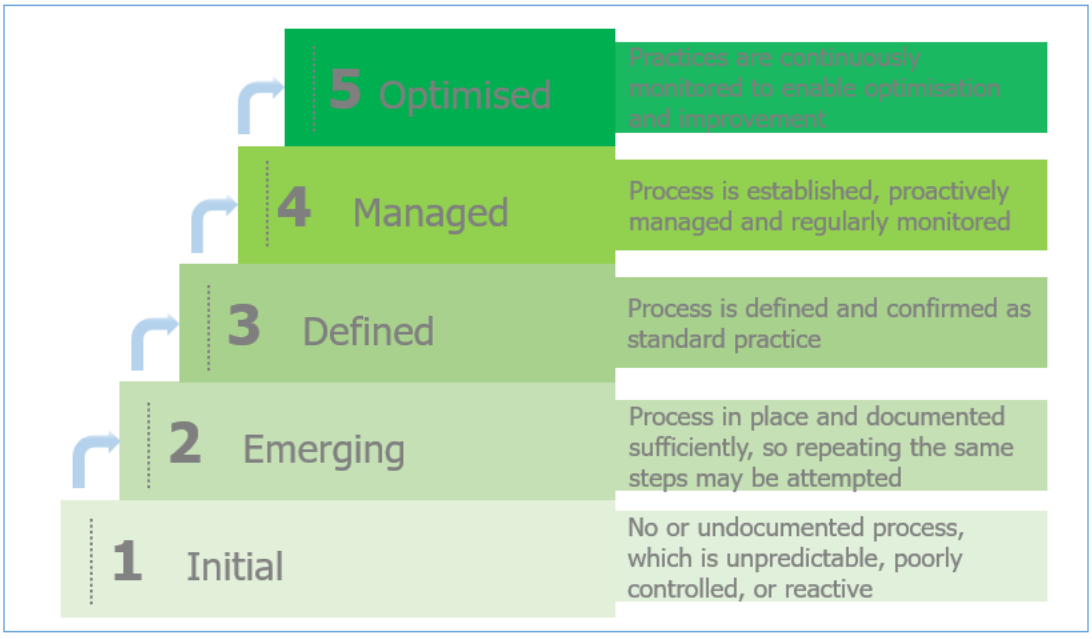
The IMMAT differs from the SAT in that it acts as a benchmarking tool providing useful information for both data holders, data users and policy makers. It is a high-level assessment of information management practices generating an overall score of information management maturity (see Table 2). It determines maturity levels across all twelve national standards and five levels of maturity based on the Capability Maturity Model (CMM)[9] and also adapted by the QUANTUM project,[*10,11] (see Figure 1). The IMMAT is therefore an easy-to-use tool for organisations to complete, when used in conjunction with the standards and associated implementation plan. Data holders answer 12 questions by selecting one of five maturity level statements that best describes how well their current practices align with each standard, providing an overall score between 12 and 60 (see Figures 2 and 3 and Table 2). Therefore, it further helps to improve information management practices enabling data holders to assess changes over-time within their own organisation and across similar organisations nationally. This will have significant benefits in terms of targeting priorities, identifying exemplars at a national level, sharing knowledge on implementation through a peer support network, and monitoring interventions over-time.

Directly linking the IMMAT to the national standards supports the broader objectives of the EHDS, by facilitating the development of capability and capacity on the appropriate management of information at every level within the system. This will provide valuable information to prepare for sharing of data for secondary purposes, fulfilling the requirements for the implementation of the EHDS and enhancing maturity of information management in a country where information and digital maturity remain low compared to other European countries. Enhancing maturity and skills is a critical component to support the scale of digitalisation and change required in Ireland over the coming years. The implementation of the national standards and the assessment of maturity will also support other outputs from the HealthData@IE grant such as the nHDC, the DAAMS, the SPE, and data quality enhancement in line with European requirements such as reporting the Data Quality and Utility Label (DQ&UL). The national standards and IMMAT include 12 standards, one of which focuses specifically on data quality, so both tools are aligned and complement each other. HIQA is also developing National Guidance for enhancing data quality and a tool to support data holders on the assessment of data quality in line with the DQ&UL, as well as associated

[*] QUANTUM aims to develop recommendations on a common label system that can be used across Member States to report on the quality and utility of datasets so that data users can identify high-quality data for different purposes. The EHDS Regulation will enforce the application of the label for datasets made available through a HDAB.

training modules. The IMMAT remains high-level so best practice will be guided by the training materials developed to support data holders which can be amended overtime and in line with the detail in Implementing Acts. In addition, HIQA is developing a Compliance Assessment Framework for Data Quality (CAF-DQ) which is also aligned with the dimensions in the DQ&UL helping to validate information provided through the DQ&UL and more broadly through the IMMAT scores provided for the data quality standard. The IMMAT findings will also subsequently be used by HIQA to develop a prioritisation schedule for compliance reviews and targeted reviews of data quality. By supporting the development of more mature data quality systems, data holders will be better prepared to ensure their compliance with the national standards and meet their obligations under the EHDS Regulation, associated Implementing Acts, and Ireland's Health Information Act.[12]

Figure 1. Definition of information management maturity levels, adapted from QUANTUM



Source: Adapted from QUANTUM Deliverable 1.2 Specification for the assessment of data holders´ maturity; November 2024. Available from: https://quantumproject.eu/wp-content/uploads/2024/12/D1.2_Specification_assessment_data_holders_maturity-submitted-1.pdf. Accessed on: 9 January 2026

Figure 2. Example of level statements from the IMMAT

Standard 1.1 We uphold people's rights relating to information

We have effective arrangements in place to ensure a person's rights under relevant legislation are promoted and upheld. We balance these rights against fundamental rights[2], human rights[3], and legitimate, public and vital interests. We are transparent about how and why we collect, use and share information.

Thinking about your organisation in relation to the above standard, please select a score based on the statement you most agree with:

Level 1 Initial	Level 2 Emerging	Level 3 Defined	Level 4 Managed	Level 5 Optimised
We have no arrangements in place to ensure people's rights relating to information are upheld.	We recognise the importance of upholding people's rights relating to information but practices remain ad hoc.	We are driving forward initiatives to uphold people's rights relating to information but the approach is not fully comprehensive and/or implemented across our organisation.	We have comprehensive arrangements in place to uphold people's rights relating to information. It is clear how and why the organisation collects, uses and shares information and we implement this practice throughout our organisation.	We have prioritised upholding people's rights within our organisation by taking a strategic and proactive approach. Our organisation promotes an inclusive, sensitive and equitable approach to information management and we actively collaborate with key stakeholders to support us in assessing and developing our practices regarding upholding rights.

Please select your score for Standard 1.1: **Level** Choose an item.

Figure 3. Example of IMMAT scoring across 12 standards and overall score

Score by Standard

Principle	Standard	Your score	Maximum score
A human rights-based approach	Standard 1.1 We uphold people's rights relating to information	2	5
	Standard 1.2 We protect people's privacy and confidentiality	4	5
Safety and wellbeing	Standard 2.1 We optimise the accessibility, use and value of information	4	5
	Standard 2.2 We undertake effective stakeholder engagement	4	5
Responsiveness	Standard 3.1 We align with best practice regarding data standards and agreed definitions	1	5
	Standard 3.2 We enhance data quality	2	5
	Standard 3.3 We ensure data security	3	5
	Standard 3.4 We develop staff capability and capacity for information management	4	5
Accountability	Standard 4.1 We develop strong organisational governance, leadership and management	2	5
	Standard 4.2 We implement strategy for information management	3	5
	Standard 4.3 We promote effective performance assurance and risk management	1	5
	Standard 4.4 We ensure compliance with relevant legislation and codes of practice	2	5
	Overall score:	32	60
	The level of information management maturity in your organisation is:	DEFINED	

Table 2: IMMAT total maturity scores

Total score	Your organisation's overall maturity
12	Initial (Level 1)
13-24	Emerging (Level 2)
25-36	Defined (Level 3)
37-48	Managed (Level 4)
49-60	Optimised (Level 5)

2. Introduction

2.1 Purpose of the IMMAT pilot

The overall aim of the IMMAT pilot is to test the IMMAT with data holders in Ireland.

The specific objectives of the IMMAT pilot were to:

1. Ensure that the tool is **usable, accessible and fit-for-purpose**. The applicability and usability of the tool in various settings were investigated and any potential gaps or challenges to implementation identified.
2. Provide an opportunity to engage with data holders to explore their preferences for the **design, format and content** of the IMMAT report.

2.2 Purpose of the report

The purpose of this report is to outline the key findings of the IMMAT pilot.

These findings will be used to:

- inform future iterations of the IMMAT, including a digital version of the tool, before it is disseminated to all data holders in Ireland
- identify any unmet needs and any areas where data holders need supports
- provide input on useful metrics for benchmarking purposes in the future.

3. Methodology

3.1 Participants

The IMMAT was piloted with four health data holders across various settings and levels of maturity, representing major national data collections/health information systems in Ireland. In early 2025, HIQA conducted a survey of data holders' levels of preparedness for the establishment of HDAB services in Ireland, called the Readiness Assessment.[13] A sample of data holders involved in the Readiness Assessment were invited to participate in the IMMAT pilot, representing four different levels of self-reported information management maturity identified through the Readiness Assessment.

This pilot represents the initial engagement with data holders to develop the IMMAT. Therefore, at this stage of piloting, a qualitative approach to data collection was deemed most appropriate. A detailed 'think-out-loud' study with a sample of major national data collections in Ireland at varying levels of maturity provided important insights into the development of a tool to ensure that it is practical and fit-for-purpose. The next phase of development involves building the IMMAT into a digital tool (Milestone 8.19 and Deliverable 8.3, October 2026) and completing a mid-scale roll-out collecting the IMMAT from high-priority data holders (Deliverable 8.4, October 2027).

3.2 Methods

For the pilot, each participating data holder was invited to take part in an individual 90-minute 'think-out-loud' session, which was conducted as a semi-structured interview. Through email, participants were sent an editable MS Word version of the IMMAT and background briefing documents outlining the purpose and key themes for discussion, at least two weeks in advance of these sessions. Data holders were asked to review the IMMAT before the session and discuss any feedback during the session.

The 'think-out-loud' sessions were conducted online using Zoom and were guided by a member of the HIQA team using a semi-structured topic discussion guide. The sessions were recorded and transcribed for note-taking purposes. The topic discussion guide was developed with input from the Work Package 8 Project team and was organised into four areas:

- data holder preparedness

- design, format and content of the IMMAT
- implementation of the IMMAT at the given organisation
- IMMAT results and outputs.

The pilot participants were invited to discuss their experience of using the tool, and each standard was discussed in detail to elicit any specific challenges or facilitators to each self-assessment question and to get a deeper insight into the completion and applicability of the IMMAT within their organisation.

In addition, participants were asked about the outputs of the pilot, for example, how data holders would like to see their scores, information management maturity reports tailored to their individual organisation, and useful benchmarking metrics.

3.3 Analysis

Content analysis was conducted using a framework analysis approach.[14] Transcripts and audio recordings from the four 'think-out-loud' sessions were reviewed, and a thematic framework was constructed using the themes outlined in the discussion guide.

Descriptive codes were generated directly from the content and any important quotes were identified. Subthemes were created based on recurring categories or sentiments that emerged as the interview data was analysed.

To ensure analytical rigour, a second iteration of this process was performed by another team member, with re-review of transcripts to identify any discrepancies. Differences in interpretation were discussed and resolved through collaborative discussion in order to reach a consensus, and uncertainties were discussed with a third team member.

The final synthesis and interpretation involved considering each theme and subtheme in the context of the whole set of interviews.

4. Key findings

4.1 Summary of key findings

The aim of the IMMAT pilot was to test the IMMAT with data holders in Ireland and specifically to gather feedback on the design, format, content and implementation of the tool, as well as ensuring that the tool is usable, accessible and fit-for-purpose.

Overall, pilot participants deemed the IMMAT a useful tool. It identifies areas for data holders to prioritise in refining their own information management maturity and was viewed as a good exercise to complete periodically in order to track improvements. It was also noted that the IMMAT could assist in the provision of key information to senior management, which in turn could help steer focus towards specific areas of information management enhancement.

One notable finding of the IMMAT pilot was that participants from organisations with more than one dataset need the flexibility of completing the assessment at both organisational and dataset level, as maturity scores may vary considerably between different datasets within the same organisation.

With regards to the design and content of the IMMAT, participants from organisations with multiple datasets noted that the ability to duplicate responses, where appropriate, would be useful. A digital tool that allows for collaboration with multiple internal users would also benefit organisations, as it was found that for the majority of participants, team involvement was required to complete the IMMAT.

In terms of the content of the assessment, many participants expressed a need for examples of each maturity level across all of the standards, they also highlighted several areas for refining the text and language. This included ensuring that the tool is applicable at both organisational and dataset level.

Participants expressed a strong interest in the tool's benchmarking functionality and said it would be useful to compare to a national average or organisations of a similar type or size but stressed that reporting would need to be anonymous.

Potential challenges to implementation identified by participants included limited time and capacity of employees. Some suggestions to support data holders included

establishing communities of practice or peer-support networks to facilitate shared learning, and trainings and or webinars to support the roll-out of the IMMAT.

To support implementation across organisations, participants advised promoting stakeholder engagement with the national standards and emphasising the tool's usefulness in helping organisations to meet their obligations under the EHDS Regulation.

4.2 Design, format and content of the IMMAT

4.2.1 Design and format

In general, data holders participating in the pilot found the IMMAT design and format "easy to navigate" and "user friendly". Some participants printed the assessment as they found the MS Word version of the tool "not the easiest" to navigate.

For larger organisations with multiple datasets, it was noted that it would be "very helpful" to be able to duplicate responses across datasets where applicable, as some organisations have datasets at different levels of maturity and would therefore have to complete numerous forms for an accurate assessment. In particular, one participant commented that this would facilitate prompt completion and return of the IMMAT: "Yeah, that would be very helpful [duplicating responses], having to re-enter it all every time would be… well, you would find that people would… be late, in getting them back in for you."

With regards to the format of the assessment, participants suggested that for future digital versions of the tool, a downloadable version would be useful to allow cross-comparison of data across datasets and facilitate internal discussion with other team members.

Participants also noted it was essential that multiple users from an organisation have access to the digital IMMAT, with one person responsible for submission. The ability to save responses and come back to it later was seen as useful for future digital versions of the IMMAT.

4.2.2 Language

Across the statements within each section of the IMMAT, participants suggested updates to its wording to improve clarity and help differentiate between different

maturity levels. In some cases, the language used was viewed as being either too vague or too specific for organisations to implement effectively.

Similarly, there was a suggestion that the language used throughout the IMMAT could be refined to accommodate a broader range of dataset types.

4.2.3 Content

The IMMAT instructions on how to complete the assessment and how the overall information management maturity score is calculated by the tool was noted to be "clear" to the pilot participants and no gaps were noted in the IMMAT content.

There was a suggestion to include guidance for data holders in the instructions section, on how to identify the best person within their organisation to complete the IMMAT. The instructions need to be clear that the IMMAT should be completed by a person(/s) familiar with the dataset being assessed and that it should be shared with colleagues across the relevant areas of the organisation, for input into the IMMAT's completion.

Clear examples for the maturity levels within each section of the IMMAT is something that pilot participants would find useful. Some participants noted that it was often difficult to choose between maturity levels as there is an element of interpretation involved. So, clear examples may enable a more consistent interpretation across organisations.

Originally, the following question was included at the start of the IMMAT: 'Overall, how would you indicate the level of maturity of your organisation with regard to information management?' The intention was to collect an organisation's initial assessment of its maturity level before completing the IMMAT and compare this to its calculated maturity score at the end of the IMMAT. However, some participants found this confusing and suggested removing this question or providing clearer instructions to explain the purpose of this question. Data holders need clarity over whether this question should be answered at an organisational level or a dataset level.

It was suggested that data holders should be more strongly encouraged to familiarise themselves with the national standards before completing the IMMAT. One participant noted: "I do think maybe…if it was more strongly worded that the user should reference the national standards for information management that might be useful. I know it's referenced, but I just feel it might be good to have a kind of a stronger

[statement], 'please look at this in conjunction with…' maybe, or it might just provide a bit of clarity for the user… but it depends on their level of knowledge". It was also suggested that reference to the standards in the aims and purpose of the IMMAT may help data holders to "focus their mind" when completing the assessment.

One participant mentioned that data holders also align with FAIR data standards,[15] and suggested that this is reflected in the IMMAT.

4.3 Completing the IMMAT

4.3.1 Complete at organisation or dataset level?

A recurring theme in the interviews was whether the assessment should apply at an organisational level or be limited to an individual dataset level. This was intentionally left undefined during the design phase of the IMMAT, to gather participants' input on which level would be more appropriate. There was a split among participants regarding whether the IMMAT should be completed at organisation or dataset level.

Some participants commented that topics such as 'governance', 'training' and 'accountability' made more sense to assess at an organisation level. Conversely, participants from organisations with multiple datasets struggled to assign accurate maturity scores, as high performing datasets tended to offset weaker ones, resulting in scores being "almost cancelled out" and masking problem areas. It is therefore clear that the IMMAT must be flexible enough to accommodate both levels of assessment in future.

4.3.2 Who is responsible?

Across organisations, a range of different roles were involved in completing the IMMAT. However, most contributors held responsibilities related to data governance or data quality.

Depending on the size and setting of an organisation, in some cases several people within the pilot participants' organisations had input into completing the IMMAT. While participants noted that the information required to complete the assessment was readily available, the time to complete the IMMAT varied considerably. This variation was largely influenced by the time spent discussing and debating scores within teams when a group approach was used and their familiarity with the national standards.

It was also noted that some organisations may have external contractors or collaborators with responsibility for certain aspects of information management, and so the organisation may also need to consult with these external bodies when completing the IMMAT, consequently affecting the time taken to complete the assessment.

4.4 Implementing the IMMAT

4.4.1 Benefits of completing the IMMAT

Participants agreed that the IMMAT aligns with their organisations' goals and recognised the benefit of completing it. In terms of encouraging other data holders to engage with the IMMAT process, participants suggested that the benefits of completing the IMMAT are clearly communicated, such as enabling data holders to prepare for their obligations under the EHDS Regulation and compliance with the national standards.

Additionally, some participants noted the usefulness of the IMMAT in highlighting gaps and areas where data holders need additional resources to assist senior management. This in turn would support organisations in reaching compliance with the national standards.

4.4.2 Potential challenges and facilitators for data holders

Participants noted that good documentation by the data holder would help expedite completion of the IMMAT. They also agreed that, as users become more proficient and the national standards become more embedded in practice, the time required to complete the IMMAT would likely decrease.

Another facilitator suggested by the pilot participants was to establish a community of practice or peer-support network among data holders, for example, a regular online meeting for staff occupying information management roles at different organisations to come together, share ideas and learn from each other.

Time and capacity constraints were cited as challenges to completing the IMMAT by most pilot participants. For example, one participant noted: "I can see pushback… it's the time to do it when we are already, you know, flat out with meeting our obligations… of getting reports out." However, the benefit of completing the IMMAT was recognised: "I think that… it would be more resource-intensive but the utility of the output I think would be greater".

In some organisations, data holders need to factor in time to allow all staff contributing to the IMMAT to read and familiarise themselves with the national standards. Training events, webinars and "easy-to-digest" videos and infographics were seen as useful accompaniments to the tool, to help get staff up to speed in this area.

Pilot participants also noted that the IMMAT will require organisations to adapt and harmonise their existing data quality processes, to accommodate interoperability and data linkage, which will also be needed in preparation for the establishment of HDAB services. Finally, some concerns were raised over how the information from the IMMAT might be used in the future and some participants noted that HIQA should be clear about how the results will be used.

4.4.3 Opinions on IMMAT outputs

The outputs of the IMMAT (for example an organisation's overall maturity score, and maturity scores for each standard and principle), were deemed to be clear and useful by all pilot participants and allowed data holders to see which areas needed improvement.

Participants suggested setting up an implementation plan in order to guide data holders on the next steps needed to improve information management maturity in their organisation. However, participants were informed that such an improvement plan already exists for the national standards and so an additional implementation plan for maturity may cause confusion.

The ability to benchmark scores against a national average or against organisations of a similar type or size in the future was also seen as useful. However, any such benchmarking would need to be anonymised, and participants urged caution when looking at maturity at a regional level as there is a lot of variation across the health sector that must be taken into account.

In addition, one participant noted the need for further caution around benchmarking and associated unintended consequences. For example, if organisations are scoring consistently higher than others they may become complacent. However, in general, participants recognised that European developments in this area should promote motivation for continuous improvements.

5. Further considerations

While this study provides valuable insights, the findings are based on the perspectives of four data holders. The detailed 'think-out-loud' qualitative approach involving a sample of major national data collections in Ireland at varying levels of self-reported information management maturity, provides important insights and was deemed the most appropriate approach at this stage of the project, given that the tool is still under development. Further engagement through an expert working group, user testing of the digital tool with data holders, and the roll-out of the IMMAT will provide additional information to inform the tool's continued development (see Section 5.2 Next steps and **Appendix C** for comprehensive information regarding the different stages of development).

5.1 Key changes to the IMMAT based on findings

Following a detailed analysis of the interview data, several key areas were identified for consideration in the enhancement of the tool.

Table 3. Key considerations for IMMAT enhancement

Suggested change to IMMAT	Detail
IMMAT questions changes	▪ Review and refine the language used for individual standards' statements. ▪ Consider including clear examples for each information management maturity level to support consistent interpretation or additional supporting guidance.
IMMAT instructions and accompanying guidance	▪ Provide clear guidance in the instructions on who should complete the assessment. ▪ Clarify whether the assessment should be completed at an organisational or dataset level.

	▪ Note that users should complete the IMMAT in conjunction with the *National Standards for Information Management in Health and Social Care*. ▪ Offer a clearer explanation or reconsider the initial question on 'overall maturity', including how it integrates with the rest of the assessment.
Digital IMMAT enhancements	▪ Consider providing a downloadable version of the assessment tool. ▪ Allow users to duplicate responses across datasets where appropriate, to reduce the administrative burden for organisations with multiple datasets. ▪ Allow multiple users from the same organisation to access and edit the assessment collaboratively.
IMMAT results and outputs improvements	▪ Clearly communicate how maturity scores will be used now and, in the future, ensuring data holders understand their purpose and implications. ▪ Investigate benchmarking metrics to help determine the most meaningful comparisons for data holders. ▪ Assess whether an additional implementation plan is required to provide further guidance on how organisations can improve their maturity scores.

5.2 Next steps

The next steps for the IMMAT (PDF) include incorporating the findings from the pilot into a digital version of the tool, conducting user testing of the digital tool and rolling-out the digital IMMAT with a broad group of data holders. The collection of maturity data from the IMMAT will be guided by and aligned to the National Health Dataset Catalogue (nHDC) roll-out with high-priority data holders in Ireland. The digital tool will be completed by October 2026 and the roll-out of the IMMAT is planned for 2027. This will include a much broader sample of data holders and will provide baseline quantitative data on the status of information management maturity across Ireland.

The pilot also identified areas where data holders may need support to implement the IMMAT. As a result, implementation support tools, such as webinars, infographics and simple guidance documents may be developed in order to meet this need.

Insights from the pilot and user testing phase will also be shared with all data holders in the future through peer-support networks, which are also being established as part of the HealthData@IE project.

Initial views on benchmarking were also gathered through this pilot. This provided useful input into the types of metrics that might be useful for both data holders and future HDAB services, which will be incorporated into the benchmarking functionality of the IMMAT.

5.3 Conclusion

The findings of the IMMAT pilot provide constructive feedback, which will inform the development of the digital tool, benchmarking functionality and future training and support needs. Learnings will be useful in the roll-out of the IMMAT to all data holders in Ireland in the future.

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Appendix A – Steering Group membership for HealthData@IE

Name	**Organisation** ***Title, Division***
Emer Doyle	Department of Health (Chair) *Principal Officer, Health Information Policy*
Rachel Flynn	Health Information and Quality Authority (HIQA) *Director, Health Information and Standards*
Barbara Foley	HIQA *Deputy Director, Health Information Quality and Assurance*
Kevin O'Carroll	HIQA (WP8) *Programme Manager, Health Information Standards*
Brendan Brady	Department of Health *Health Information Policy*
Bríd Burke	Health Research Consent Declaration Committee
Frances Burns	Northern Ireland Trusted Research Environment
Helen Conroy	Department of Health *Health Information Policy*
Eamonn Coyne	Department of Health *eHealth and Health Information Systems*
Steve Dempsey	Irish Cancer Society *Patient Representative*
Kevin Devlin	National Research Ethics Committee *HRB PPI Reviewer, Patient Representative*
Eoin Farrell	Department of Health *Health Information Policy*
Laura Flannelly	Department of Health *Statistics and Analytics Unit*
Lorna Gaffney-White	Department of Health *Health Information Policy*
Richard Greene	Health Service Executive (HSE) *Chief Clinical Information Officer*
Loretto Grogan	HSE Office of the Chief Clinical Officer *Chief Information Officer – Nursing and Midwifery*
Prof Markus Helfert	National University of Ireland, Maynooth
Noreen Hynes	Health Service Executive (HSE) *Chief Data and Analytics Office*

Name	**Organisation** ***Title, Division***
Ailish Kelly	Department of Health *Health Information Policy*
Kathryn Kissane	Health Service Executive (HSE) *Deputy Delivery Director, Standards & Shared Care Records*
Anthony Macken	Central Statistics Office (CSO) *Researcher Coordination Unit*
Dr Teresa Maguire	Health Research Board (HRB) *Director of Research Strategy and Funding*
Denise Manton	National University of Ireland, Maynooth
Elizabeth McCrohan	Department of Health *Statistics and Analytics Unit*
Elaine Murray	EIT Health (Ireland-UK) *Public Affairs*
Cáit Ní Chorcora	Department of Health *Health Information Policy*
Azul O'Flaherty	Department of Health *Health Information Policy*
Ronan O'Kelly	Department of Health *eHealth and Health Information Systems*
John O'Neill	Department of Health *Research Policy and Innovation Unit*
Róisín O'Neill	Department of Health *Research Policy and Innovation Unit*
Christopher Ryan	Department of Health *Research Services Unit*
Kerry Ryder	Health Service Executive (HSE) *Research and Development*
Niall Sinnott	Department of Health *eHealth and Health Information Systems*
Clodagh Thorne	Department of Health *Health Information Policy*
Caitriona Wray	Department of Health *Health Information Policy*

Appendix B – Working Group membership for Work Package 8

Name	**Organisation** ***Title***
Barbara Foley	Health Information and Quality Authority (HIQA, Chair) *Deputy Director, Health Information Quality and Assurance*
Maria Ryan	HIQA (WP8) *Programme Manager, Health Information Quality and Assurance*
Kevin O'Carroll	HIQA (WP8) *Programme Manager, Health Information Standards*
Anthony Macken	Central Statistics Office (CSO) *Researcher Coordination Unit*
Brendan Brady	Department of Health, Health Information Policy Unit *Data Officer & Health Literacy Policy Lead*
Cliona O'Donovan	National Office of Clinical Audit (NOCA) *Head of Data Science*
Frank Moriarty	Royal College of Surgeons in Ireland (RCSI) *Senior Lecturer*
Gabriela Avram	HSE National Patient Forum *Patient partner*
Helen Conroy	Department of Health, Health Information Policy
Jean Kelly	National Cancer Registry Ireland (NCRI) *Data Quality Manager*
Leonard Browne	University of Limerick (UL) *Senior Research Fellow (Biostatistics)*
Loretto Grogan	HSE Office of the Chief Clinical Officer *Chief Information Officer – Nursing and Midwifery*
Lorraine McNerney	Department of Public Expenditure, NDP Delivery and Reform - Office of the Government Chief Information Officer *Head of Data Governance*
Louise Cullen	HSE Health Protection Surveillance Centre *Principal Epidemiologist*
Niall Halliday	HSE Office of the Chief Clinical Officer *National Chief HSCP Clinical Information Officer*
Patricia Wilson	HSE Office of the Chief Clinical Officer *ICT Project Manager, Shared Care Record*
Sarah Craig	Health Research Board (HRB) *Director of Health Information and Evidence*

Appendix C – How the IMMAT was developed

In developing the IMMAT, an initial review of maturity models internationally was undertaken to identify any similar assessment tools and maturity frameworks. It was agreed to develop the IMMAT based on the Capability Maturity Model (CMM).[9] The CMM supports structured growth by helping to systematically enhance data management practices. It has been adapted for various domains such as, cyber security, service delivery processes and European health information projects such as QUANTUM.[10,11] Following the initial scoping review, the project team drafted the maturity tool aligning it with the principles and standards from the *National Standards for Information Management in Health and Social Care* as outlined in Table 1.

This involved mapping these standards against the five levels of maturity within the CMM (see Figure 1) and the four compliance levels used by the national standards (compliant, substantially compliant, partially compliant and non-compliant). A new level descriptor (Level 5) was created for organisations who perform beyond compliance and continuously optimise and improve their information management maturity. This tool was reviewed internally within HIQA, before sharing it with the Working Group for Work Package 8. A workshop was organised to gather detailed feedback from the Working Group members which included the perspective of patient representatives, data users, policy makers and academic experts. Changes were made to the IMMAT based on this feedback before progressing the tool through HIQA internal governance checks and additional updates were made at each stage of feedback.

In September 2025, a pilot of the IMMAT was conducted with four major data holders in Ireland. The holders were invited to take part in an individual 90-minute 'think-out-loud' session, which was conducted as a semi-structured interview. Data holders were asked to review the IMMAT before the session and discuss any feedback during the session. The 'think-out-loud' sessions were conducted online using Zoom and were guided by a member of the HIQA team using a semi-structured topic discussion guide. The pilot participants were invited to discuss their experience of using the tool, and each standard was discussed in detail to elicit any specific barriers or facilitators to each self-assessment question and to get a deeper insight into the completion and applicability of the IMMAT within their organisation. In addition, participants were asked about the outputs of the pilot, for example, how data holders would like to see their scores, information management maturity reports tailored to their individual organisation, and useful benchmarking metrics. The results were analysed and a report

of findings was written. The suggested changes to the content of the IMMAT were considered by the team and incorporated into an updated version of the IMMAT.

In 2026, an external solution provider was contracted to develop a digital version of the IMMAT with benchmarking functionality. This will allow data holders to easily self-assess their level of information management maturity online. This digital IMMAT will be tested with a group of end users to provide a data holder perspective and identify any usability issues with and enhancements to the online version of the tool. The IMMAT is being developed in tandem with the nHDC, sharing the same platform and login credentials, driving further efficiencies in the system and ensuring seamless accessibility for data holders. The digital IMMAT will be completed by October 2026 and the roll-out of the IMMAT is planned for 2027.

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